



Defect-mediated hydroxylation of multi-walled carbon nanotubes as metal-free catalysts to enhance catalytic performance for oxidative dehydrogenation of ethylbenzene using CO₂

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ABSTRACT

The fabrication and hydroxylation of surface defects on multi-walled carbon nanotubes (MWCNTs) are performed through a single or combined method involving hydrothermal, ultrasonic and ball-milling treatments. These treatments cannot change the tubular morphology of the graphite structure in MWCNTs but they can alter the surface defect structure on MWCNTs. Surface defects on MWCNTs can promote the hydroxylation of the MWCNTs to produce hydroxyl groups. However, these defects are difficult to hydroxylate when occupied by other functional groups, such as carboxyl groups, even though the hydroxylation processes are the same. After hydroxylation, the MWCNTs with surface defects can be employed as metal-free catalysts for the oxidative dehydrogenation of ethylbenzene using CO₂ as soft oxidants. The inherent disorder or artificial skeleton defects on MWCNTs cannot directly act as catalytic active sites, but they can be hydroxylated to enhance the catalytic activity of the MWCNTs. For the same type of MWCNT defects formed after hydroxylation, the catalytic activity is closely related to the defect density. If the disorder or skeleton defects cause a higher defect density on the hydroxylated MWCNTs, the MWCNTs will exhibit a higher catalytic activity. Due to the different structures and positions of the MWCNT defect carbons, the artificial skeleton defects on MWCNTs more efficiently promote the hydroxylation of MWCNTs compared to the disorder defects, decreasing their defect density and enhancing their catalytic activity. Therefore, MWCNTs-BOH36 with skeleton defects has a catalytic activity that is twice as high as that of MWCNTs-CS-120 with disorder defects.

1. Introduction

Currently, the commercial production of styrene primarily employs the catalytic dehydrogenation of ethylbenzene (EB). However, this process requires the vaporization of a large amount of water into steams which results in high energy consumption and high production costs. In order to solve this problem, a gas-phase oxidant is used to replace water vapor and to perform catalytic oxidative dehydrogenation (ODH) of EB to styrene. CO₂, a greenhouse gas, has become one of the most promising candidates for use as an oxidant in the ODH of EB to styrene [1,2]. This reaction is a low energy consumption, low production cost and environmentally compatible process. Theoretically and experimentally, many materials are able to perform the ODH of EB to styrene using CO₂ as a soft oxidant [1–3]. Interestingly, the low-cost and high-performance catalytic materials with weak anti-oxidizing abilities are

developed in the catalytic ODH system. Carbon-based materials are typical and excellent catalytic materials that have the advantage of environmental compatibility and are fabricated from carbonaceous chemical substances and sustainable biological resources. These materials may be used as important catalytic components to catalyze the ODH of EB in CO₂ and obtain interesting catalytic properties [4,5]. Therefore, carbon-based materials are very promising catalytic materials for the ODH of EB to styrene using CO₂ as a soft oxidant.

As fascinating carbon-based materials, multi-walled carbon nanotubes (MWCNTs), have unique mechanical, electrical and thermal properties and have many potential applications in energy conversion/storage, biotechnology, electronic nanodevices, wastewater treatment and heterogeneous catalysis [5–9]. MWCNTs have numerous surface structural defects due to large-scale growth processes via chemical vapor deposition (CVD). Structural defects of MWCNTs damage the sp²-

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